Gabriele Ciccolella



MSc Graduate in Intelligent Systems Engineering
Cybersecurity & Quantum Security | Python Development | AI
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Portfolio

PROFILE

MSc graduate with 110/110 cum laude and project experience in Python development, AI and microservice architectures. Developed a BB84 quantum key distribution simulator for my master's thesis, combining cybersecurity knowledge with hands-on software engineering.

EDUCATION

MSc in Intelligent Systems Engineering 10/2024 - 10/2026
Campus Bio-Medico University of Rome **Final grade: 110/110 cum laude**

BSc in Industrial Engineering 09/2021 - 12/2024
Campus Bio-Medico University of Rome **Final grade: 104/110**

SELECTED PROJECTS

Quantum Key Distribution (QKD/BB84) MSc Thesis
- Designed and implemented an 8-microservice BB84 simulator in Python, FastAPI and Docker Compose, covering error estimation, reconciliation and key verification.
- Analysed how channel losses and errors affect protocol decisions and final key availability.

AI Agent and Microservice Architecture Academic Project
- Built a requirements-analysis agent with n8n, Ollama and Redis; integrated 6 Docker Compose services with metrics, dashboards and availability alerts.

Photovoltaic Power Forecasting Deep Learning Project
- Implemented a PyTorch encoder-decoder LSTM using 72 hours of history and known future weather to forecast the next 24 hours; applied feature engineering, early stopping and baseline comparison using MAE/RMSE.

Machine Learning Studies Academic Projects
- Earthquake damage prediction: Random Forest/LightGBM, Optuna and nested cross-validation.
- Rehabilitation exercise recognition: compared 8 classifiers.
- Oncology data classification: implemented a KNN model.

Zanzo Multiplayer Card Game | Demo Personal Project
- Developed a Python/JavaScript app for 2-8 players with PBKDF2 password hashing, token authentication, rate limiting, SQLite and WebSocket; automated tests and Cloudflare Workers deployment.

TECHNICAL SKILLS

Software & testing: Python, JavaScript, SQL, HTML/CSS, FastAPI, Pydantic; Git, pytest, Playwright.
Systems & DevOps: REST APIs, microservices, Docker, Docker Compose, Kubernetes; Prometheus, Grafana, GitHub Actions.
AI & machine learning: PyTorch, scikit-learn, LightGBM, Optuna, pandas, NumPy; n8n, Ollama, Redis.
Cybersecurity: Ethical hacking labs: Kali Linux, Wireshark, Nmap, Burp Suite; TCP/IP, OSI, authentication, authorisation and hashing.
Quantum security: QKD/BB84, QBER estimation, finite-sample bounds, error reconciliation and KMS availability analysis.
Robotics (academic): Manipulator kinematics, geometric Jacobians and trajectory planning.

SKILLS & INTERESTS

Languages: Italian (native) | English (B2) | Spanish (basic)
Office: Microsoft Excel, Word and PowerPoint.
Interests: Chess (three-time university tournament finalist) | Competitive football (S.S. Lazio).

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